Transitivity of derivability in the sequent calculus

The Open Logic Project contributors

openlogic:first-order-logic/sequent-calculus:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both hypotheses are used only through their finite witnesses; the proof displays one bussproofs prooftree with the cut inference

Editorial note. The label prop:transitivity names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Cut rule. From derivations of Gamma sequent A and of A, Delta sequent B one derives Gamma, Delta sequent B by cut.. The displayed derivation composes pi_0 (Gamma_0 sequent A) over pi_1 (A, Delta_0 sequent B) by cut.
- Finite-subset characterization of derivability. Gamma proves A iff some finite Gamma_0 subseteq Gamma has Gamma_0 sequent A derivable.. Both hypotheses are unpacked into finite witnesses Gamma_0 and Delta_0 with derivations pi_0 and pi_1.

Proposition 0.1 (Transitivity)**.** *If* $\Gamma \vdash A$ *and* $\{A\} \cup \Delta \vdash B$*, then* $\Gamma \cup \Delta \vdash B$*.*

Proof. If $\Gamma \vdash A$, there is a finite $\Gamma_0 \subseteq \Gamma$ and derivation π_0 of $\Gamma_0 \Rightarrow A$. If $\{A\} \cup \Delta \vdash B$, then for some finite subset $\Delta_0 \subseteq \Delta$, there is derivation π_1 of $A, \Delta_0 \Rightarrow B$. Consider the following derivation:

$$\dfrac{\begin{array}{cc}\vdots\,\pi_0 & \vdots\,\pi_1 \\ \Gamma_0 A & A, \Delta_0 B\end{array}}{\Gamma_0, \Delta_0 B}\ \text{Cut}$$

Since $\Gamma_0 \cup \Delta_0 \subseteq \Gamma \cup \Delta$, this shows $\Gamma \cup \Delta \vdash B$. □